



# Tanushka Das

Roll no:- 240123066

B.Tech- Mathematics And Computing

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## EDUCATION

| Degree/Certificate | Institute/Board                                 | CGPA/Percentage | Year         |
|--------------------|-------------------------------------------------|-----------------|--------------|
| B.Tech. Major      | Indian Institute of Technology, Guwahati        | 8.10(Current)   | 2024-Present |
| Senior Secondary   | Telangana State Board of Intermediate Education | 98.7%           | 2024         |
| Secondary          | Board Of Secondary Education,Telangana          | 10.00           | 2022         |

## PROJECTS

- Corporate Transportation Optimization System (Vehicle Routing Engine)** *Januray 2026 - March2026*  
*Kriti-inter hostel Technical Competition* *GitHub*
  - Developed a vehicle routing optimization engine using K-Means clustering, greedy insertion, repair heuristics, and 2-opt local search to solve constrained routing problems.
  - Co-designed and implemented the backend logic in Python, handling route assignment, constraint validation (time windows, capacity, preferences), and route optimization.
  - Reduced route cost by 81–95%** while maintaining operational constraints, applying algorithm design and performance optimization techniques.
- CVM++: Custom Compiler & Stack-Based Virtual Machine** *April 2026 - May 2026*  
*Coding Club* *GitHub*
  - Built a **custom compiler and stack-based virtual machine in C++**, implementing lexical analysis, recursive-descent parsing, AST construction, and semantic processing.
  - Developed a **custom bytecode format** and compiler backend for translating source programs into executable instructions.
  - Engineered a stack-based interpreter supporting variable management, expression evaluation, and complete source-to-execution workflow.
- AI powered Resume Analyzer and Job tracking Application** *April 2026 - May 2026*  
*Coding Club* *CHNAGE GITHUB*
  - CHANGE ALL CONTENT** Built a custom compiler and stack-based VM in C++, implementing the complete source-to-execution pipeline.
  - Parsing:** Developed a handwritten lexer, recursive-descent parser, and AST generator with operator-precedence handling.
  - Compilation:** Designed a custom bytecode format and compiler backend for code generation.
  - Engineered a stack-based interpreter supporting variable management, expression evaluation, and complete source-to-execution workflow.
- DISRUPT Startup Bootcamp Website** *July. 2025*  
*Ecell IIT Guwahati* *Website | GitHub*
  - Collaborated with the WebOps team to develop and maintain the **official DISRUPT startup bootcamp website** for E-Cell IIT Guwahati.
  - Built frontend features and improved user experience for a production platform serving **70+** startups and **25+** mentors.

## TECHNICAL SKILLS

- Programming:** C, C++,JavaScript, Java\*
- Web Technology::** HTML,CSS,GSAP\*, React ,Node Js, Express Js
- Databases::** MongoDB, MYSQL
- Tools/Frameworks:** Git, Github, REST APIs, Postman
- Operating Systems:** Windows, Linux\* *\* Elementary proficiency*

## KEY COURSES TAKEN

- Mathematics:** Linear Algebra, Basic Calculus, Discrete Maths, Probability & Random Processes, Real analysis,Design and Analysis of Algorithms,Financial Engineering\*,Data structures and Algorithms
- Computer Science::** Digital Design, Database Management System, Computer Organization and Architecture *\*Awarded AA grade*

## ACHIEVEMENTS

- Secured 1st position in the "Zero to One"**,Secured 1st place among competing teams in the “Zero to One” startup challenge organized by E-Cell IIT Guwahati by developing a business strategy and investor pitch for an EV logistics startup. *2025*

## EXTRACURRICULARS

- Senior Executive Disrupt,ECELL Club, IIT Guwahati**,Managed startup bootcamps at Disrupt, coordinating with startups across India. *May 2025-Jan 2026*
- Public Relations Associate,Alcheringa, IIT Guwahati**,Served as a Public Relations Associate, strengthening communication, stakeholder engagement, and event management skills. *Nov. 2024 - Mar-2025*